

EXAM 7 – SPRING 2015

25. (2 points)

a. (0.5 point)

Define each of the following:

- i. Insurance underwriting risk
- ii. Liquidity risk

b. (0.75 point)

Identify an event scenario that illustrates comovement of insurance underwriting risk and liquidity risk. Briefly describe how the potential for this scenario could affect the assessment of these two risk types.

c. (0.75 point)

Identify a risk mitigation strategy for the scenario identified in part b. above. Briefly describe how this strategy would reduce each component of risk considered here.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 25	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: C6: Describe the rationale for, methods for, and effect of managing insurance and financial risk.
SAMPLE ANSWERS	
Part a: 0.5 point	
<i>Sample Answer 1</i> <u>Underwriting risk</u> is the risk associated with writing insurance. It encompasses many things, including product design risk, inadequate reserve/premium risk, accumulation risk, catastrophe risk, policyholder behavior risk, etc. The crux is having inadequate premium to cover the exposure. <u>Liquidity risk</u> is the risk of having insufficient liquid assets to be able to meet a sudden cash demand. <i>Sample Answer 2</i> <u>Underwriting risk</u> is the risk that we are not writing business at the profit we need to obtain. Could be due to pricing inadequacies or mix of business issues. <u>Liquidity risk</u> is the risk that we will not have enough funds to cover our liabilities that become due, and cannot meet current obligations.	
Part b: 0.75 point	
<i>Sample Answer 1</i> A hurricane hitting the East Coast is the scenario we will consider. For a property insurer, this will create a big cash demand, as there will be many policyholders affected and claims will increase. Claims are part of underwriting risk. The company may have more than enough assets to meet these payments, but if the assets are not liquid or cannot be converted to cash at their full value, then there is a liquidity risk and the company may not be able to pay their policyholders within a reasonable timeframe. <i>Sample Answer 2</i> A major economic downturn or financial crisis ○ Policyholders for certain lines of business (e.g. Worker's Compensation) may submit many more claims than expected, and the higher frequency is not considered in the pricing, leading to an underwriting loss. ○ The higher-than-expected claims volume leads to a demand for payment in a shorter period of time, possibly forcing the company to liquidate assets at a discount <i>Sample Answer 3</i> Insurance industry becomes aware of latent claim risk, such as asbestos ○ Newly-discovered cause of claims was not accounted for when policies were priced and sold, leading to an underwriting loss ○ Many claims caused by this latent risk source are reported in a short period, leading to a large demand for payment from the insurance company, which is forced to liquidate assets at a discount to meet demands.	
Part c: 0.75 point	

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Sample Answer 1

One mitigation strategy would be to purchase catastrophe reinsurance. This would reduce underwriting risk by reducing the potential for losses due to a catastrophic event. It reduces liquidity risk since the reinsurer will pay some of the losses and the primary will not need to come up with cash as quickly (spreads risk over time by paying reinsurance premiums).

Sample Answer 2

The company could issue a catastrophe bond. In the event of a large hurricane/earthquake, the investors' principal covers a significant portion of the loss, reducing the underwriting risk. It provides an influx of liquid assets to address the heightened demand for claim payment.

EXAMINERS' REPORT

In general, candidates performed fairly well on this question, though parts b and c were more challenging than part a. Some candidates did not fully understand how liquidity risk arises from the event scenarios they described, and they frequently had difficulty coming up with a risk mitigation strategy to reduce liquidity risk.

Part a

Candidates were expected to know the definitions of underwriting and liquidity risk. Candidates generally performed very well on this question, particularly the definition of underwriting risk. The most common error was stating that liquidity risk arises from "insufficient assets" without specifying that those assets must be liquid. Some candidates referred to liquidating assets before their maturity; however, the real risk is not being able to sell the assets for their full value.

Part b

Candidates were expected to identify a single event that would affect both underwriting and liquidity risk. Most candidates were able to identify such an event, but some struggled to adequately explain how the event would impact each type of risk. In attempting to describe the impact, some candidates failed to describe how the event would cause an increased demand for liquid assets.

Part c

Candidates were expected to provide a risk mitigation strategy to reduce the two types of risk. This was the most challenging part of the question. Most candidates provided a strategy, but sometimes the strategy only addressed one of underwriting and liquidity risk without addressing the other.